

# Free Value Or Else

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## Abstract

This paper proposes free function overloads of `value_or` — and the related `reference_or`, `or_invoke`, and `or_construct` — over any nullable type, such as `std::optional`, `std::expected`, and pointers to objects. They compute the type of the result correctly, via `common_type` and `common_reference`, as the member `std::optional::value_or` could not. At LEWG’s request it also carries the member functions of [P3413R0] — `value_or_construct` and `value_or_else` on `optional` and `expected` — so that the two proposals are reviewed, and can be adopted, as one.

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## Changes Since Last Version

- **R1** Unified with [P3413R0] at LEWG’s request: Corentin Jabot joins as co-author, and the member functions `value_or_construct` and `value_or_else` are folded into the proposal and wording.
- **R0** Initial revision.

## 1 Comparison table

### 1.1 Raw pointer with a manual null check

The fundamental nullable type is the raw pointer, and the fundamental idiom for providing a default is the conditional expression guarded by a null check. The free function names the idiom, removes the repetition of the checked entity, and computes the type of the result correctly.

```
Cat* cat = find_cat("Fido");           Cat* cat = find_cat("Fido");
return cat ? *cat : Cat{"Default"};    return std::value_or(cat, Cat{"Default"});
```

### 1.2 The value\_or truncation pitfall

Because `std::optional<T>::value_or` converts its argument to `T`, a perfectly reasonable-looking sentinel silently truncates. The free function computes the common type of the contained value and the alternative, and converts both *outward* to that type instead of forcing the alternative *inward* through `T`.

```
std::optional<std::uint16_t> o = get();    std::optional<std::uint16_t> o = get();
auto v = o.value_or(-1);                  auto v = std::value_or(o, -1);
// v is uint16_t{65535}                    // v is int; -1 is preserved
```

### 1.3 Reference alternative without copies

When both the contained value and the alternative are references to long-lived objects, the result should be a reference. Member `value_or` cannot do this; `reference_or` can, and statically rejects the cases where the result would dangle.

```
std::optional<Logger&> opt = get_logger();    std::optional<Logger&> opt = get_logger();
Logger& l = opt ? *opt : default_logger();    Logger& l = std::reference_or(
                                              opt, default_logger());
```

## 2 Motivation

`std::optional<T>::value_or` answers the most common question asked of an optional — “the value, or this instead” — and answers it with the wrong type. It returns `T`, and it requires the alternative to be convertible to `T`. Every use therefore funnels both the contained value and the alternative through `T`, whether or not `T` is the right type for the result. The consequences are familiar: sentinels truncate (`optional<uint16_t>.value_or(-1)` is 65535), wider alternatives narrow, and reference results are simply impossible, so `value_or` forces a copy even when both candidate results are references to long-lived objects.

This is not fixable in place. Changing the return type of a member function that has been shipping since C++17 is both an API and an ABI break, and the `&&`-qualified overload set compounds the problem. During the work on `std::optional<T&>` [P2988R12] the question was reopened in earnest, because for an optional reference the cost of the wrong answer is no longer a pessimization but a wrong program: returning `T` from `optional<T&>::value_or` copies, returning `T&` dangles when the alternative is a temporary. After extensive discussion in LEWG, P2988 concluded that “there is no particularly wonderful solution for `value_or` that does not involve a time machine,” adopted the least objectionable member behavior — return `remove_cv_t<T>` by value — and explicitly forecast this paper: free functions over all types modeling an optional-like concept, where the return type can finally be computed correctly.

Existing practice diverges precisely because there is no one member-function answer that satisfies everyone:

| Implementation | Behavior                                                                    |
|----------------|-----------------------------------------------------------------------------|
| Standard       | <code>optional&lt;T&gt;::value_or</code> returns a <code>T</code>           |
| Boost          | <code>optional&lt;T&gt;::value_or</code> returns a <code>T</code>           |
| TartanLlama    | <code>optional&lt;T&amp;&gt;::value_or</code> returns a <code>T&amp;</code> |
| Flux           | <code>optional&lt;T&gt;::value_or</code> returns a <code>T</code>           |
| Think-Cell     | returns the type of the ternary, similar to <code>common_reference</code>   |
|                | returns <code>common_reference</code> , with some caveats                   |

The machinery to compute the right answer postdates `optional`. `std::common_type` existed but was not used; `std::common_reference` arrived with ranges in C++20; and `std::reference_constructs_from_temporary_v` [P2255R2] arrived in C++23, making it possible for the first time to *statically reject* exactly the dangling cases that made reference-returning `value_or` untenable. The conditional expression `b ? *m :`

u has computed the common type all along; the library can now name that computation, add the lifetime check the core language does not perform, and do so once, for every nullable type.

That last point is the generalization this paper takes from [P1255R13]: being “contextually convertible to bool and dereferenceable” is a protocol, not a property of `std::optional`. Raw pointers, `std::shared_ptr`, `std::unique_ptr`, `std::expected<T,E>`, and `std::optional<T>` all speak it. Each of them today requires either a hand-written conditional or a type-specific idiom to express “the value, or this instead.” A single constrained free function family covers them all, with one set of semantics to learn and one place to get the corner cases right.

### 3 Design

The proposal is four free function templates over a nullable concept:

```
template <class T>
concept nullable = requires(const T t) {
    bool(t);
    *(t);
};

template <nullable T, class U,
         class R = std::common_reference_t<std::iter_reference_t<T>, U&&>>
constexpr auto reference_or(T&& m, U&& u) -> R;

template <nullable T, class U,
         class R = std::common_type_t<std::iter_reference_t<T>, U&&>>
constexpr auto value_or(T&& m, U&& u) -> R;

template <nullable T, class I,
         class R = std::common_type_t<std::iter_reference_t<T>,
                                     std::invoke_result_t<I>>>
constexpr auto or_invoke(T&& m, I&& invocable) -> R;

template <class Ret = void, nullable T, class... Args,
         class R = std::conditional_t<std::is_void_v<Ret>,
                                     std::remove_cvref_t<std::iter_reference_t<T>>, Ret>>
constexpr R or_construct(T&& m, Args&&... args);

template <class Ret = void, nullable T, class E, class... Args,
         class R = std::conditional_t<std::is_void_v<Ret>,
                                     std::remove_cvref_t<std::iter_reference_t<T>>, Ret>>
constexpr R or_construct(T&& m, std::initializer_list<E> il, Args&&... args);
```

All four have the same engaged-path behavior — produce `*m`, converted to `R` — and differ in how the alternative is supplied (a value, a value bound as a reference, a nullary invocable, constructor arguments) and in how `R` is computed (`common_type`, `common_reference`, or the element type itself).

#### 3.1 The nullable concept

The concept is the “maybe protocol” of [P1255R13]: an object is nullable if it is contextually convertible to bool and dereferenceable. The requirements are stated on `const T`: observation must not require mutable access. A type whose `operator bool` or `operator*` is non-const does not model `nullable`; asking “the value or a default” is a query, and a type that cannot be queried through const access is not answering that question.

Types verified to model the concept: `std::optional<T>`, `std::expected<T,E>`, raw object pointers `T*`, `std::shared_ptr<T>`, `std::unique_ptr<T>`, and `std::optional<T&>` as specified by [P2988R12]. Types verified not to model it: arithmetic types and bool (contextually convertible to bool but not dereferenceable), `std::string` and the containers (neither), `std::nullptr_t`, `void`, and types providing only one of the two operations. Iterators in general do not model `nullable` because they are not contextually convertible to bool — correctly so, since an iterator’s validity is not a property the iterator itself can report.

One refinement remains open. P1255’s `nullable_object` additionally requires that the result of dereferencing be an equality-preserving *object*, which excludes function pointers; the minimal concept above admits them

(a function pointer is contextually convertible to `bool` and dereferenceable), and they are then rejected only downstream, when `common_type` fails. LEWG guidance is sought on whether the concept should require `is_object_v` of the referent, rejecting function pointers at the constraint rather than in the return-type computation.

### 3.2 Exposition-only, or a named public concept?

Whether `nullable` is an exposition-only constraint on these three functions or a named, public concept is a LEWG question in its own right, and the paper presents it for a poll.

The case for a public `std::nullable`: the protocol is vocabulary. [P2988R12] forecast these functions “over all types modeling optional-like, `concept std::maybe`”, and [P1255R13] is titled, in part, “a concept to constrain maybes” — the concept has been treated as a deliverable in its own right throughout this line of work. Users writing their own functions over nullable types — a `value_or` variant with different policy, an algorithm taking a range of nullables — need the same constraint, and without a public name each codebase re-derives it, not always identically. A named concept can be tested directly (`static_assert(std::nullable<T>)`), appears by name in diagnostics, and fixes the meaning of “optional-like” across the ecosystem once.

The case for exposition-only: a named standard concept is forever, and this one is not yet stable. The open refinements above — the `is_object_v` question, const-observability, and the semantic requirements a named concept would have to state in words (dereference is valid only when the contextual conversion yields `true`; observation is equality-preserving) — can be adjusted release to release while the concept is exposition-only, and cannot be once user code constrains against the name. The library has a directly analogous precedent: *boolean-testable*, among the most-used constraints in the standard, is deliberately exposition-only for exactly this reason. With [P1255R13] concluded (see the discussion of `yield_if` below), this paper is the concept’s sole remaining home; no cross-paper coordination is needed, but there is also only one chance to name it right.

The asymmetry favors caution: an exposition-only concept can be promoted to a public name in a later standard without breaking anything, while a public name can never be demoted or meaningfully tightened. R0 therefore specifies `nullable` as exposition-only, with promotion — under whatever name LEWG prefers — as the poll alternative.

### 3.3 Return type computation

`std::iter_reference_t<T>` is `decltype(*declval<T>())` — the type produced by dereferencing an lvalue of the nullable type. For every nullable above with an `int` payload it is `int&`; for a `const optional<int>` it is `const int&`; for `const int*` it is `const int&`. The return type is then the common type (for `value_or` and `or_invoke`) or the common reference (for `reference_or`) of that dereference type and the alternative.

For `value_or`, `common_type` decays, so the result is a prvalue, and mixed-type uses promote instead of truncating:

| nullable T                                                                             | alternative U              | result R            |
|----------------------------------------------------------------------------------------|----------------------------|---------------------|
| <code>optional&lt;int&gt;</code>                                                       | <code>int</code>           | <code>int</code>    |
| <code>expected&lt;int,E&gt;, int*, shared_ptr&lt;int&gt;, unique_ptr&lt;int&gt;</code> | <code>int</code>           | <code>int</code>    |
| <code>optional&lt;int&gt;</code>                                                       | <code>long</code>          | <code>long</code>   |
| <code>optional&lt;int&gt;</code>                                                       | <code>double</code>        | <code>double</code> |
| <code>optional&lt;uint16_t&gt;</code>                                                  | <code>int</code> (e.g. -1) | <code>int</code>    |
| <code>optional&lt;string&gt;</code>                                                    | <code>string</code>        | <code>string</code> |

This is the type the equivalent conditional expression would have produced, which is exactly the point: `value_or(m, u)` is the library spelling of `m ? *m : u`, not of `m ? *m : T(u)`.

### 3.4 Value categories, and observing rather than consuming

As specified, dereference of the nullable is always performed on an lvalue: `iter_reference_t` is computed from `T&`, and the engaged path returns `static_cast<R>(*m)`. An rvalue nullable does *not* move its payload out: `value_or(std::move(opt), u)` copies the contained value where `std::move(opt).value_or(u)` would move it. The functions observe; they do not consume.

This is a deliberate simplification for R0, and the author seeks LEWG guidance. The alternative — computing the dereference type as `decltype(*std::forward<T>(m))` so that rvalue optionals move — recovers the member function’s efficiency for the rvalue case, at the cost of making the return type depend on the value

category of the nullable and of introducing a moved-from path that is harder to reason about under the dangling analysis of `reference_or`. The simpler rule has a simple statement: the result is what the lvalue conditional expression would have produced. Performance-sensitive consuming uses retain the member functions.

### 3.5 `reference_or`

`reference_or` computes `common_reference_t<iter_reference_t<T>, U&&>` and, when that type is a reference, binds the result to the contained value or to the alternative without copying either. Const qualification flows through the computation as it should: a `const optional<int>` or a `const int&` alternative each force `const int&` as the result; an `int*` with an `int&` alternative yields `int&`, writable through, and mutation through the result reaches the original object — the function introduces no copies.

Testing surfaced a subtlety: the common reference of two unrelated lvalue references need not be a reference. `common_reference_t<int&, long&>` is `long`, and `common_reference_t<string&, const char(&)[2]>` is `string`; in such calls `reference_or` compiles, the dangling guards are vacuously satisfied — `R` is not a reference, so nothing can bind to a temporary — and the function quietly returns a prvalue, behaving exactly like `value_or`. No lifetime hazard arises, and this is the conditional expression’s own behavior faithfully reproduced; but a caller who wrote `reference_or` for its reference semantics receives a copy. LEWG guidance is sought on whether `reference_or` should additionally mandate that the computed type *is* a reference type, making the degenerate calls ill-formed and steering them to `value_or`.

The danger of returning references is binding one to a temporary, and that is now statically detectable. The implementation carries two guards:

```
static_assert(!std::reference_constructs_from_temporary_v<R, U>);
static_assert(!std::reference_constructs_from_temporary_v<R, T&>);
```

The first rejects the alternative materializing a temporary: in `reference_or(opt, 0)` the computed common reference of `int&` and `int&&` is `const int&`, which would bind to a temporary materialized from the literal, and the call is ill-formed. The second rejects the symmetric case where the conversion of `*m` to `R` would materialize a temporary. Together they admit exactly the calls whose result refers to an object that outlives the expression, and reject at compile time the calls that core language lifetime rules would quietly accept and then dangle.

Whether these should be `static_asserts` or constraints is a design question for LEWG. As `static_asserts`, a dangling call is a hard, clearly-diagnosed error wherever it occurs — including inside an overload-resolution context, where it is an error rather than a removed candidate. As constraints, the overload disappears SFINAE-style, which composes better with generic code probing for usability but turns a lifetime bug into a silent “no matching function” or, worse, a fallback to a different overload. The author’s position is that a dangling `reference_or` call is a bug to be reported, not a candidate to be quietly discarded, and R0 proposes the `static_assert` behavior (specified as *Mandates*).

The name is also open. The Tokyo discussion of the P2988 homework concluded that `value_or` is not an available name for a reference-returning function, and suggested `ref_or`; this paper spells it `reference_or`, matching the library’s general preference for unabbreviated names, and will bring a naming table for a poll.

### 3.6 `or_invoke` and laziness

`value_or` and `reference_or` evaluate their alternative unconditionally — it is an ordinary function argument. When the alternative is expensive to construct, that cost is unwelcome on the engaged path, and the established cure is to pass a nullary invocable that is called only when needed. `or_invoke(m, f)` returns `*m` converted to `R` when `m` is engaged and `f()` converted to `R` otherwise, with `R = common_type_t<iter_reference_t<T>, invoke_result_t<I>>`; the invocable is not called on the engaged path.

Because `or_invoke` uses `common_type`, an invocable returning a reference still produces a value: `common_type_t<int&, int&>` is `int`. The family therefore covers three of four quadrants — eager value (`value_or`), eager reference (`reference_or`), lazy value (`or_invoke`, and `or_construct` below) — and deliberately omits the fourth. A lazy reference-returning form is specifiable (the dangling guard would apply to `invoke_result_t<I>` just as it does to `U`), but the use case — a reference alternative that is expensive to *name*, rather than to construct — is thin, and it is not proposed. If LEWG wants the quadrant filled, it composes naturally as `reference_invoke` later.

The obvious name, `or_else`, is taken: `std::optional::or_else` returns an `optional`, re-wrapping the result, and giving the same name to a value-producing free function would make `x.or_else(f)` and `or_else(x, f)` mean different things. [P3413R0] faces the same collision for its member function and chooses `value_or_else`. For the free function this paper proposes `or_invoke`, and will poll the name alongside `value_or_else` and `or_eval`.

### 3.7 `or_construct`: the fallback as constructor arguments

`or_construct` is the free-function analogue of [P3413R0]’s member `value_or_construct`: the alternative is supplied not as a value but as constructor *arguments*, and the fallback object is constructed lazily, only on the disengaged path — on the engaged path nothing is constructed from them at all. An `initializer_list` overload provides the same convenience the member form has.

Its return-type rule is deliberately different from the other three. There is no fallback *value* whose type could participate in a `common_type` computation — only arguments waiting to become one — so by default `R` is the decayed element type itself, `remove_cvref_t<iter_reference_t<T>>`, and the conversion runs *inward*: the arguments construct an `R` directly, and the engaged payload converts to `R`. This is the member function’s model, and here it is the right one; the outward common-type computation is meaningless without a second value.

What the free function can offer that the member cannot is choice of the constructed type. The leading `Ret` template parameter (defaulted through a `void` sentinel) lets the caller name the result type explicitly:

```
std::optional<std::pmr::string> m = lookup(key);
auto s = std::or_construct<std::string>(m, "default");
// std::string either way: converted from the payload, or
// constructed from the arguments --- chosen by the caller
```

The fallback is direct-initialized, so `explicit` constructors are usable — the arguments were always explicitly a construction request. Like `value_or` and `or_invoke`, `or_construct` returns a prvalue and cannot dangle; with `optional<T&>` the result is a value copy of the referent or a freshly constructed object, never a reference.

### 3.8 Why free functions

The member-function route is closed twice over. For `std::optional` and `std::expected` the existing `value_or` cannot change its return type, and a second member with corrected semantics would have to coexist with the first forever, on each class, kept in sync by hand. For pointers — raw and smart — there are no members to add at all. A constrained free function template is the only form that can state the semantics once and apply them to every type speaking the protocol, including user-defined nullables, which model the concept simply by being contextually convertible to `bool` and dereferenceable.

These are ordinary function templates, not customization point objects. There is nothing to customize: the semantics are fully determined by the protocol, and a type that wants different behavior for “value or default” is not modeling `nullable`, it is doing something else. The usual argument for a CPO — dispatching to a type’s own implementation — is therefore absent. Whether the names should nonetheless inhibit ADL is left to LEWG.

### 3.9 `std::expected`, and the error

`std::expected<T,E>` models `nullable`: both `bool(e)` and `*e` are valid on a `const expected`. Consequently `value_or(e, u)` works, and discards the error, exactly as the member `value_or` does. This is by design: the moment the question is “the value, or this instead,” the error has been decided to be uninteresting. Uses that need the error have `error_or` and the monadic interface on the member side; no error-observing overload is proposed here.

### 3.10 What about `yield_if`, and the end of P1255

[P2988R12] forecast a fourth function, `yield_if`, producing a view of zero or one elements, alongside the three proposed here. It was to remain with the [P1255R13] ranges work — but that work is concluded. Range support for `std::optional` [P3168R2] supplanted `views::maybe`, and `views::nullable` with it: an `optional` now *is* the view of zero or one elements, and a pointer-like `nullable` reaches the same place through `optional<T&>` [P2988R12], as `p ? optional<T&>(*p) : nullopt`. The author does not expect to revise P1255 further; this paper inherits its remaining deliverables, the concept and the free functions over it.



That covers `yield_if` as well: `yield_if(cond, value)` is now spellable as `cond ? optional(value) : nullopt`, which is itself a range. What residual demand exists is for a named convenience in comprehension-style pipelines; if implementation experience shows it is wanted, it is a small separate paper. It is explicitly not part of this proposal.

## 4 Relation to P3413R0

[P3413R0] (“A more flexible `optional::value_or` (else!)”), reviving [P2218R0], proposes member functions `value_or_construct` — constructing the fallback `T` lazily and in place from forwarded arguments, with `initializer_list` overloads — and `value_or_else` — taking a lazily evaluated nullary callable — on both `std::optional` and `std::expected`. The two papers aim at the same member function and fix different defects, and LEWG should see them together.

P3413 fixes the *cost* of the alternative: with `value_or_construct` the fallback object is not constructed at all on the engaged path, and `value_or_else` defers arbitrary computation. It deliberately does not touch the return type — everything still returns `T`, with the alternative still required to convert to `T` — so the truncation pitfall, the forced copies, and the impossibility of reference results all remain. It is also necessarily member-by-member: `optional` and `expected` each grow six functions, and pointers of any kind get nothing. On `optional<T&>` it explicitly defers to P2988.

This paper fixes the *type* of the answer and the *reach* of the idiom: results are computed via `common_type/common_reference` rather than funneled through `T`, reference results become possible and dangling-checked, and one set of free functions covers every nullable type. Both lazy cases now overlap. `or_invoke` and member `value_or_else` differ in form (free vs. member), reach (any nullable vs. `optional/expected`), and return type (`common_type` vs. `T`). `or_construct` and member `value_or_construct` share the in-place, lazy construction model and the `initializer_list` overload; the free function additionally lets the caller name the constructed type (`or_construct<string>(m, args...)`) and works on every nullable.

|                                     | P3413R0                                       | this paper                                               |
|-------------------------------------|-----------------------------------------------|----------------------------------------------------------|
| form                                | members                                       | free functions                                           |
| applies to                          | <code>optional</code> , <code>expected</code> | any nullable (incl. pointers)                            |
| return type                         | <code>T</code>                                | <code>common_type</code> / <code>common_reference</code> |
| truncation pitfall                  | unchanged                                     | fixed                                                    |
| reference results                   | no                                            | <code>reference_or</code> , dangling-checked             |
| lazy alternative                    | <code>value_or_else</code>                    | <code>or_invoke</code>                                   |
| in-place fallback construction      | <code>value_or_construct</code>               | <code>or_construct</code> , callable-chosen result type  |
| rvalue nullable moves payload       | yes ( <code>&amp;&amp;</code> overloads)      | no (observes only; see Design)                           |
| <code>optional&lt;T&amp;&gt;</code> | deferred to P2988                             | covered by the concept                                   |

The proposals are not alternatives to choose between. LEWG in Brno asked for a single proposal — “Can we ask you and Corentin to come back with one proposal?” — and Corentin is amenable; this chapter, and the wording below, are that unification. The two compose. Adopt the members and `optional` and `expected` gain the lazy, in-place fallback, spelled the way a member is spelled and moving out of an rvalue. Adopt the free functions and the corrected semantics — the computed result type, reference results, the reach to pointers and every other nullable — become available generically, over those same types and over the ones that have no members to grow. A caller writing `x.value_or_construct(args...)` and one writing `or_construct<R>(m, args...)` are served by the same design at two spellings, and neither forecloses the other. This paper therefore proposes both: the free functions of the preceding chapters, and the members of [P3413R0], whose wording it reproduces below.

## 5 Principles for Reification of Design

The same principles that governed [P2988R12] govern here.

**Errors that can be detected at compile time must be detected at compile time.** The concept rejects non-nullable arguments at the constraint; `reference_or` rejects every call whose result would bind to a temporary via `reference_constructs_from_temporary_v`, on both the alternative path and the contained-value path.

**Never knowingly produce a dangling reference.** The functions take no address and extend no lifetime; `reference_or`’s result always refers to an object that existed before the call. The rejected cases are precisely those where the core language would have materialized a temporary: a prvalue alternative whose common reference with the dereference type is a const reference (`reference_or(opt, 0)`), and conversions of the contained value to the result type that materialize.

**Value-producing functions never dangle by construction.** `value_or` and `or_invoke` return prvalues of a decayed common type; there is no reference in their result to check.

These principles are enforced by a negative-compilation test suite in the reference implementation: each forbidden combination is a translation unit that must fail to compile, with the diagnostic matched against the constraint or `static_assert` that is required to fire.

## 6 Implementation Experience

The reference implementation is `beman.free_value_or` [Downey\_free\_value\_or]. The implementation floor is C++23, required for `reference_constructs_from_temporary_v`; the functions are otherwise C++20 material.

The test suite exercises the matrix of: nullable types `std::optional<T>`, `std::expected<T,E>`, `T*`, `std::shared_ptr<T>`, `std::unique_ptr<T>`, and (gated on a C++26 toolchain, via the Beman `optional` reference implementation of P2988 [The Beman Project\_beman\_optional]) `std::optional<T&>`; engaged and disengaged; the nullable as lvalue, const lvalue, and rvalue; the alternative as lvalue and rvalue; mixed-type combinations; constant evaluation of all four functions; laziness of `or_invoke` and `or_construct` and eagerness of `value_or`; for `or_construct`, construction arities from zero arguments through multi-argument emplacement and the `initializer_list` overload, with construction confirmed to occur only on the disengaged path; and negative-compilation cases for both concept violations and dangling rejection. Reference identity and mutation round-trips confirm that `reference_or` introduces no copies: writes through its result reach the contained object when engaged and the alternative when not. The `optional<T&>` suite additionally verifies rebinding semantics: after an `optional<T&>` is rebound to a new referent, `value_or` produces and `reference_or` refers to the new referent, and disengaging falls through to the alternative.

The suite — 106 tests at this writing, all green — found no defects in the implementation. It did surface two behaviors reported in Design: that `reference_or` degrades to a prvalue result when the common reference of unrelated types is not a reference type, with the dangling guards then vacuous; and that `or_invoke`’s use of `common_type` decays reference-returning invocables to values, leaving the lazy-reference quadrant of the family deliberately empty. The negative-compilation harness was itself validated: each WILL-FAIL translation unit was checked to fail for the required reason by mutating its expected-diagnostic pattern and observing the test fail.

Verified `common_reference` results for `reference_or`, showing const propagation:

| nullable (lvalue)                           | alternative (lvalue)        | result                      |
|---------------------------------------------|-----------------------------|-----------------------------|
| <code>optional&lt;int&gt;&amp;</code>       | <code>int&amp;</code>       | <code>int&amp;</code>       |
| <code>const optional&lt;int&gt;&amp;</code> | <code>int&amp;</code>       | <code>const int&amp;</code> |
| <code>optional&lt;int&gt;&amp;</code>       | <code>const int&amp;</code> | <code>const int&amp;</code> |
| <code>expected&lt;int,int&gt;&amp;</code>   | <code>int&amp;</code>       | <code>int&amp;</code>       |
| <code>int*</code>                           | <code>int&amp;</code>       | <code>int&amp;</code>       |
| <code>shared_ptr&lt;int&gt;&amp;</code>     | <code>int&amp;</code>       | <code>int&amp;</code>       |
| <code>optional&lt;string&gt;&amp;</code>    | <code>string&amp;</code>    | <code>string&amp;</code>    |

The current implementation, in full:

```
template <class T>
concept nullable = requires(const T t) {
    bool(t);
    *(t);
};

template <nullable T, class U,
        class R = std::common_reference_t<std::iter_reference_t<T>, U&&>>
constexpr auto reference_or(T&& m, U&& u) -> R {
```



```

    static_assert(!std::reference_constructs_from_temporary_v<R, U>);
    static_assert(!std::reference_constructs_from_temporary_v<R, T&>);
    return bool(m) ? static_cast<R>(*m) : static_cast<R>((U&&)u);
}

template <nullable T, class U,
         class R = std::common_type_t<std::iter_reference_t<T>, U&&>>
constexpr auto value_or(T&& m, U&& u) -> R {
    return bool(m) ? static_cast<R>(*m) : static_cast<R>(std::forward<U>(u));
}

template <nullable T, class I,
         class R = std::common_type_t<std::iter_reference_t<T>,
                                     std::invoke_result_t<I>>>
constexpr auto or_invoke(T&& m, I&& invocable) -> R {
    return bool(m) ? static_cast<R>(*m) : static_cast<R>(invocable());
}

template <class Ret = void, nullable T, class... Args,
         class R = std::conditional_t<std::is_void_v<Ret>,
                                     std::remove_cvref_t<std::iter_reference_t<T>>, Ret>>
constexpr R or_construct(T&& m, Args&&... args) {
    return bool(m) ? static_cast<R>(*m) : R(std::forward<Args>(args)...);
}

template <class Ret = void, nullable T, class E, class... Args,
         class R = std::conditional_t<std::is_void_v<Ret>,
                                     std::remove_cvref_t<std::iter_reference_t<T>>, Ret>>
constexpr R or_construct(T&& m, std::initializer_list<E> il,
                        Args&&... args) {
    return bool(m) ? static_cast<R>(*m) : R(il, std::forward<Args>(args)...);
}

```

The member side is implemented too. `value_or_construct` and `value_or_else` were added to the Beman reference implementations of `optional` [The Beman Project `beman_optional`] and `expected` [The Beman Project `beman_expected`] — vendored into `beman.free_value_or` and edited in place — on both the primary templates and their reference specializations. `beman.expected` already carries `expected<T&,E>`, `expected<T,E&>`, and `expected<T&,E&>`, so the members run over references now, ahead of any such paper reaching the working draft; standardizing those forms is the only dependency, not implementing them. The two libraries’ own suites — GoogleTest for `optional`, Catch2 for `expected` — run as part of this project’s tests, and cover laziness (a construction counter that never ticks on the engaged path), the `initializer_list` overload, rvalue move-out, constant evaluation, and the reference specializations. All green.

## 7 Proposal

Add to the standard library four constrained free function templates — `value_or`, `reference_or`, `or_invoke`, and `or_construct` — and the exposition-level concept that constrains them, with the semantics described above, targeting C++29. The functions should be usable on any type that is contextually convertible to `bool` and dereferenceable, including `std::optional`, `std::expected`, raw pointers, `std::shared_ptr`, `std::unique_ptr`, and `std::optional<T&>`.

Add as well the member functions of [P3413R0]: `value_or_construct` — with its `initializer_list` overload — and `value_or_else`, on both `std::optional` and `std::expected`, each in `const&` and `&&` form, producing the fallback lazily and in place. Their wording is reproduced below, unchanged in substance from P3413. `std::optional<T&>` [P2988R12] is in C++26, and the members apply to it directly, returning a value; a future `std::expected<T&,E>` would carry them the same way, and that one case depends on the expected-over-references work, as the wording notes. The two feature sets are severable: LEWG may take the free functions, the members, or both, but the point of unifying is that it need not.

Open questions on which LEWG guidance is sought before wording review:

- whether the dangling guards of `reference_or` are *Mandates* (the proposed default) or *Constraints*;

- whether `reference_or` should mandate that the computed common reference is a reference type, rather than silently degrading to a prvalue result for unrelated types (see Design);
- the names `reference_or` (vs. `ref_or`), `or_invoke` (vs. `value_or_else`, `or_eval`), and `or_construct` (vs. P3413's `value_or_construct`);
- whether the concept requires `is_object_v` of the referent, excluding function pointers at the constraint;
- whether rvalue nullables should move their payload (see Design);
- whether the concept is exposition-only (the proposed default) or named and public (see Design);
- header placement: `<optional>` is where users will look, `<utility>` matches the genericity, a new header is cleanest for freestanding; the functions should be freestanding in any case.

## 8 Wording

Initial draft wording, relative to [N5008], to be completed after LEWG design review. Placement (shown here as `<utility>`) is provisional.

Add to the header synopsis:

```
namespace std {
    template<class T>
        concept nullable = requires(const T t) { // exposition only
            bool(t);
            *(t);
        };

    template<nullable T, class U>
        constexpr common_reference_t<iter_reference_t<T>, U&&>
            reference_or(T&& m, U&& u);

    template<nullable T, class U>
        constexpr common_type_t<iter_reference_t<T>, U&&>
            value_or(T&& m, U&& u);

    template<nullable T, class I>
        constexpr common_type_t<iter_reference_t<T>, invoke_result_t<I>>
            or_invoke(T&& m, I&& invocable);

    template<class Ret = void, nullable T, class... Args>
        constexpr conditional_t<is_void_v<Ret>,
                                remove_cvref_t<iter_reference_t<T>>, Ret>
            or_construct(T&& m, Args&&... args);

    template<class Ret = void, nullable T, class E, class... Args>
        constexpr conditional_t<is_void_v<Ret>,
                                remove_cvref_t<iter_reference_t<T>>, Ret>
            or_construct(T&& m, initializer_list<E> il, Args&&... args);
}
```

```
    template<nullable T, class U>
        constexpr common_reference_t<iter_reference_t<T>, U&&>
            reference_or(T&& m, U&& u);
```

1 Let R be `common_reference_t<iter_reference_t<T>, U&&>`.

2 *Mandates:* `reference_constructs_from_temporary_v<R, U>` is false and `reference_constructs_from_temporary_v<R, T>` is false.

3 *Effects:* Equivalent to: `return bool(m) ? static_cast<R>(*m) : static_cast<R>(std::forward<U>(u));`

```
    template<nullable T, class U>
        constexpr common_type_t<iter_reference_t<T>, U&&>
            value_or(T&& m, U&& u);
```

4 Let R be `common_type_t<iter_reference_t<T>, U&&>`.

5 *Effects:* Equivalent to: `return bool(m) ? static_cast<R>(*m) : static_cast<R>(std::forward<U>(u));`

```

        template<nullable T, class I>
        constexpr common_type_t<iter_reference_t<T>, invoke_result_t<I>>
            or_invoke(T&& m, I&& invocable);
6    Let R be common_type_t<iter_reference_t<T>, invoke_result_t<I>>.
7    Effects: Equivalent to: return bool(m) ? static_cast<R>(*m) : static_cast<R>(invocable());
        template<class Ret = void, nullable T, class... Args>
        constexpr conditional_t<is_void_v<Ret>,
                                remove_cvref_t<iter_reference_t<T>>, Ret>
            or_construct(T&& m, Args&&... args);
8    Let R be conditional_t<is_void_v<Ret>, remove_cvref_t<iter_reference_t<T>>, Ret>.
9    Mandates: is_constructible_v<R, Args...> is true.
10   Effects: Equivalent to: return bool(m) ? static_cast<R>(*m) : R(std::forward<Args>(args)...);
        template<class Ret = void, nullable T, class E, class... Args>
        constexpr conditional_t<is_void_v<Ret>,
                                remove_cvref_t<iter_reference_t<T>>, Ret>
            or_construct(T&& m, initializer_list<E> il, Args&&... args);
11   Let R be conditional_t<is_void_v<Ret>, remove_cvref_t<iter_reference_t<T>>, Ret>.
12   Mandates: is_constructible_v<R, initializer_list<E>&, Args...> is true.
13   Effects: Equivalent to: return bool(m) ? static_cast<R>(*m) : R(il, std::forward<Args>(args)...);

```

## 8.1 Member functions, from P3413R0

The following member wording is reproduced from [P3413R0], relative to [N5008], and is proposed jointly with the free functions above. It is purely additive: the existing `value_or` is unchanged. Return types and *Effects* follow P3413; the *Mandates* spell the construction requirement as `is_constructible_v<T, Args...>`, matching the *Effects*.

### 8.1.1 optional [optional.optional], [optional.observe]

To the synopsis of `optional` in [optional.optional.general], after the `value_or` declarations, add:

```

template<class... Args> constexpr T value_or_construct(Args&&... args) const&&;
template<class... Args> constexpr T value_or_construct(Args&&... args) &&;

template<class U, class... Args>
constexpr T value_or_construct(initializer_list<U> il, Args&&... args) const&&;
template<class U, class... Args>
constexpr T value_or_construct(initializer_list<U> il, Args&&... args) &&;

template<class F> constexpr T value_or_else(F&& f) const&&;
template<class F> constexpr T value_or_else(F&& f) &&;

```

To [optional.observe], add:

```

        template<class... Args> constexpr T value_or_construct(Args&&... args) const&&;
1    Mandates: is_copy_constructible_v<T> is true and is_constructible_v<T, Args...> is true.
    Effects: Equivalent to: return has_value() ? **this : T(std::forward<Args>(args)...);
        template<class... Args> constexpr T value_or_construct(Args&&... args) &&;
2    Mandates: is_move_constructible_v<T> is true and is_constructible_v<T, Args...> is true.
    Effects: Equivalent to: return has_value() ? std::move(**this) : T(std::forward<Args>(args)...);
        template<class U, class... Args>
        constexpr T value_or_construct(initializer_list<U> il, Args&&... args) const&&;
3    Mandates: is_copy_constructible_v<T> is true and is_constructible_v<T, initializer_list<U>&,
    Args...> is true. Effects: Equivalent to: return has_value() ? **this : T(il, std::forward<Args>(args)...);
        template<class U, class... Args>
        constexpr T value_or_construct(initializer_list<U> il, Args&&... args) &&;

```

- 4 *Mandates:* `is_move_constructible_v<T>` is true and `is_constructible_v<T, initializer_list<U>&, Args...>` is true. *Effects:* Equivalent to: `return has_value() ? std::move(**this) : T(il, std::forward<Args>(args)...);`
- ```
template<class F> constexpr T value_or_else(F&& f) const&;
```
- 5 Let U be `invoke_result_t<F>`. *Mandates:* `is_copy_constructible_v<T>` is true and `is_convertible_v<U, T>` is true. *Effects:* Equivalent to: `return has_value() ? **this : std::forward<F>(f)();`
- ```
template<class F> constexpr T value_or_else(F&& f) &&;
```
- 6 Let U be `invoke_result_t<F>`. *Mandates:* `is_move_constructible_v<T>` is true and `is_convertible_v<U, T>` is true. *Effects:* Equivalent to: `return has_value() ? std::move(**this) : std::forward<F>(f)();`
- 7 The reference specialization `optional<T&>` [P2988R12] carries these members too; its wording is given together with `expected<T&,E>` below.

### 8.1.2 `expected` [`expected.expected`], [`expected.object.obs`]

To the synopsis of `expected` in [`expected.expected.general`], after the `value_or` declarations, add:

```
template<class... Args> constexpr T value_or_construct(Args&&... args) const&;
template<class... Args> constexpr T value_or_construct(Args&&... args) &&;

template<class U, class... Args>
constexpr T value_or_construct(initializer_list<U> il, Args&&... args) const&;
template<class U, class... Args>
constexpr T value_or_construct(initializer_list<U> il, Args&&... args) &&;

template<class F> constexpr T value_or_else(F&& f) const&;
template<class F> constexpr T value_or_else(F&& f) &&;
```

To [`expected.object.obs`], add:

- ```
template<class... Args> constexpr T value_or_construct(Args&&... args) const&;
```
- 1 *Mandates:* `is_copy_constructible_v<T>` is true and `is_constructible_v<T, Args...>` is true. *Effects:* Equivalent to: `return has_value() ? **this : T(std::forward<Args>(args)...);`
- ```
template<class... Args> constexpr T value_or_construct(Args&&... args) &&;
```
- 2 *Mandates:* `is_move_constructible_v<T>` is true and `is_constructible_v<T, Args...>` is true. *Effects:* Equivalent to: `return has_value() ? std::move(**this) : T(std::forward<Args>(args)...);`
- ```
template<class U, class... Args>
constexpr T value_or_construct(initializer_list<U> il, Args&&... args) const&;
```
- 3 *Mandates:* `is_copy_constructible_v<T>` is true and `is_constructible_v<T, initializer_list<U>&, Args...>` is true. *Effects:* Equivalent to: `return has_value() ? **this : T(il, std::forward<Args>(args)...);`
- ```
template<class U, class... Args>
constexpr T value_or_construct(initializer_list<U> il, Args&&... args) &&;
```
- 4 *Mandates:* `is_move_constructible_v<T>` is true and `is_constructible_v<T, initializer_list<U>&, Args...>` is true. *Effects:* Equivalent to: `return has_value() ? std::move(**this) : T(il, std::forward<Args>(args)...);`
- ```
template<class F> constexpr T value_or_else(F&& f) const&;
```
- 5 Let U be `invoke_result_t<F>`. *Mandates:* `is_copy_constructible_v<T>` is true and `is_convertible_v<U, T>` is true. *Effects:* Equivalent to: `return has_value() ? **this : T(std::forward<F>(f)());`
- ```
template<class F> constexpr T value_or_else(F&& f) &&;
```
- 6 Let U be `invoke_result_t<F>`. *Mandates:* `is_move_constructible_v<T>` is true and `is_convertible_v<U, T>` is true. *Effects:* Equivalent to: `return has_value() ? std::move(**this) : T(std::forward<F>(f)());`
- 7 `expected<void,E>` gains neither member — there is no value to construct — and no `error_or_construct` or `error_or_else` is proposed, following [P3413R0].

### 8.1.3 Reference specializations, and the `expected-over-references` dependency

`std::optional<T&>` [P2988R12] is in C++26; `std::expected<T&,E>` is not standardized yet, but is expected to follow. On both, these members return a value rather than a reference — exactly as `value_or` does on those

specializations — and are declared `const`, there being no rvalue overload to distinguish. For `optional<T&>` and `expected<T&,E>`, add:

```
template<class... Args>
constexpr remove_cv_t<T> value_or_construct(Args&&... args) const;
template<class U, class... Args>
constexpr remove_cv_t<T> value_or_construct(initializer_list<U> il, Args&&... args) const;
template<class F>
constexpr remove_cv_t<T> value_or_else(F&& f) const;
```

- <sup>1</sup> Let `X` be `remove_cv_t<T>`, and for `value_or_else` let `U` be `invoke_result_t<F>`. *Mandates:* `is_convertible_v<T&, X>` is true; for `value_or_construct`, `is_constructible_v<X, Args...>` (respectively `is_constructible_v<X, initializer_list<U>&, Args...>`) is true; for `value_or_else`, `is_convertible_v<U, X>` is true. *Effects:* Equivalent to, respectively:
 

```
return has_value() ? *this : X(std::forward<Args>(args)...);
return has_value() ? *this : X(il, std::forward<Args>(args)...);
return has_value() ? *this : std::forward<F>(f)();
```
- <sup>2</sup>
- <sup>3</sup> The `optional<T&>` wording above is a plain addition to a C++26 facility; nothing about it is contingent. Only the `expected<T&,E>` case is: it depends on the expected-over-references work, which is not yet in the working draft. Both reference forms are implemented already — the Beman `expected` carries `expected<T&,E>`, `expected<T,E&>`, and `expected<T&,E&>`, and the Beman `optional` the adopted `optional<T&>` — so they are exercised now, not merely specified (see Implementation Experience).

## 8.2 Feature test macro

Add a new feature test macro `__cpp_lib_value_or`, set to the date of adoption, for the free functions. For the member functions, bump `__cpp_lib_optional` and `__cpp_lib_expected` to the date of adoption, per [P3413R0]; these are independent of each other and of the `optional<T&>` macro.

## 9 Impact on the standard

A pure library extension. The free functions change no existing class or function. The member functions add `value_or_construct` and `value_or_else` to `optional` and `expected` without altering any existing member, so nothing existing changes meaning; being member function templates, they add no ABI surface. The free functions require C++23 library support (`reference_constructs_from_temporary_v`); the members impose no requirement beyond the class they are added to.

## 10 Acknowledgments

Thanks to Peter Sommerlad, co-author of [P2988R12], where the member `value_or` question was fought to a standstill, and to the LEWG reviewers of P1255 and P2988 whose objections shaped these functions. The Tokyo review homework on `value_or` alternatives surveyed the implementation divergence reported here. Thanks to Marc Mutz, whose [P2218R0] originated the member `value_or_construct` and `value_or_else` that [P3413R0] revived, and to Barry Revzin for the wording feedback carried through that paper.

## 11 Document history

- **R1** 2026-08-02 — Unified with [P3413R0] following LEWG’s Brno review, which asked for a single proposal. Corentin Jabot joins as co-author; the members `value_or_construct` and `value_or_else` are folded into the proposal, wording, and implementation.
- **R0** Brno, 2026-06 — Initial draft, reviewed by LEWG. The free-function follow-on promised in [P2988R12], concluding the line of work begun in [P1255R13].

## References

[Downey\_free\_value\_or] Stephen Downey. `beman.free_value_or`. [https://github.com/steve-downey/free\\_value\\_or](https://github.com/steve-downey/free_value_or).

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- [The\_Beman\_Project\_beman\_expected] The Beman Project. `beman.expected`. <https://github.com/bemanproject/expected>.
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